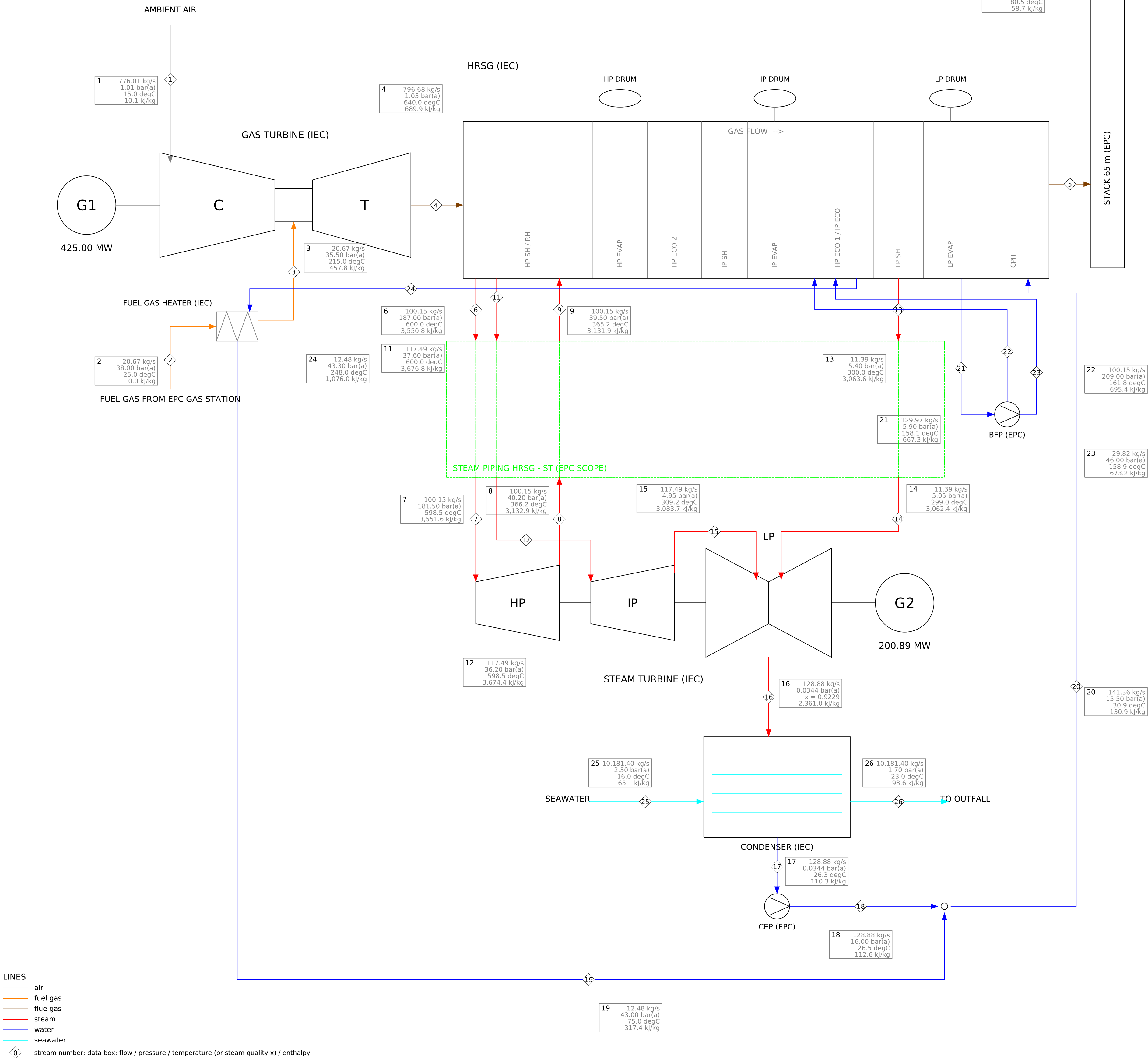




STREAM TABLE - SRC-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet (IEC terminal point)	776.01	1.01	15.0	-10.1	-7.8
2	Fuel gas at performance gas heater inlet (IEC terminal point)	20.67	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	20.67	35.50	215.0	457.8	9.5
4	GT exhaust gas to HRSG inlet	796.68	1.05	640.0	689.9	549.6
5	Flue gas at HRSG outlet to stack	796.68	1.01	80.5	58.7	46.8
6	HP steam at HRSG outlet (HP SH outlet)	100.15	187.00	600.0	3,550.8	355.6
7	HP steam at ST main stop valve inlet	100.15	181.50	598.5	3,551.6	355.7
8	Cold reheat steam at HP turbine exhaust	100.15	40.20	366.2	3,132.9	313.8
9	Cold reheat steam at HRSG reheater inlet	100.15	39.50	365.2	3,131.9	313.7
10	IP superheated steam (joins cold reheat inside the HRSG)	17.34	40.30	330.0	3,041.7	52.7
11	Hot reheat steam at HRSG outlet	117.49	37.60	600.0	3,676.8	432.0
12	Hot reheat steam at ST IP stop/control valve inlet	117.49	36.20	598.5	3,674.4	431.7
13	LP steam at HRSG outlet	11.39	5.40	300.0	3,063.6	34.9
14	LP steam at ST LP admission	11.39	5.05	299.0	3,062.4	34.9
15	IP turbine exhaust (crossover, internal)	117.49	4.95	309.2	3,083.7	362.3
16	LP turbine exhaust to condenser (UEEP)	128.88	0.0344	x 0.923	2,361.0	304.3
17	Condensate at hotwell outlet / CEP suction	128.88	0.0344	26.3	110.3	14.2
18	Condensate at CEP discharge	128.88	16.00	26.5	112.6	14.5
19	Fuel gas heater water return to condensate	12.48	43.00	75.0	317.4	4.0
20	Condensate to HRSG condensate preheater inlet	141.36	15.50	30.9	130.9	18.5
21	Feedwater from LP drum to boiler feed pumps	129.97	5.90	158.1	667.3	86.7
22	HP feedwater at BFP discharge	100.15	209.00	161.8	695.4	69.6
23	IP feedwater at BFP interstage outlet	29.82	46.00	158.9	673.2	20.1
24	IP economiser water to performance fuel gas heater	12.48	43.30	248.0	1,076.0	13.4
25	Circulating seawater to condenser inlet	10,181.40	2.50	16.0	65.1	662.9
26	Circulating seawater from condenser outlet	10,181.40	1.70	23.0	93.6	952.9

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	425.00	MW
ST output (generator terminals)	200.89	MW
Gross output	625.89	MW
Auxiliary loads	-10.32	MW
Transformer losses (GSU, UAT, MV/LV)	-1.93	MW
NET OUTPUT (HV terminals)	613.64	MW
Heat input (LHV)	981.41	MW
Gross heat rate (LHV)	5,645	kJ/kWh
NET HEAT RATE (LHV)	5,758	kJ/kWh
Net efficiency (LHV)	62.53	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	101,584	Sm3/h
Circulating water flow	36,057	m3/h

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.035 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.048 MW (tolerance 0.97 MW)
OK	energy balance FGH: residual +0.005 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.113 MW (tolerance 0.98 MW)
OK	energy balance HRSG: residual +2.057 MW vs stated casing loss 2.011 MW (tolerance 0.97 MW)
OK	energy balance PIPE: residual +0.301 MW (tolerance 1.14 MW)
OK	energy balance ST: residual -0.004 MW (tolerance 0.82 MW)
WARN	fuel LHV from the composition (ISO 6976) 47.48 MJ/kg vs DP-CASE-0018 48.4 MJ/kg (-1.90 %)
OK	auxiliary + transformer losses 1.96 % of gross vs DP-CASE-0031 <= 2.2 %
OK	condenser pressure 34.40 mbar(a) vs DP-OWNR-0021 <= 36.0 mbar(a)
OK	gas flow 101,584 Sm3/h = 88.3 % of DP-OWNR-0033 (115,000 Sm3/h)
OK	net export 613.6 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)
OK	gross output 625.89 MW vs IG-01 >= 615.00 (margin +1.77 %)
OK	gross heat rate 5,644.89 kJ/kWh vs IG-02 <= 5,735.00 (margin +1.57 %)
OK	GT output 425.00 MW vs IG-03 >= 418.00 (margin +1.67 %)
OK	condenser pressure 34.40 mbar(a) vs IG-04 <= 35.50 (margin +3.10 %)
OK	net output 613.64 MW vs PG-01 >= 600.00 (margin +2.27 %)
OK	net heat rate 5,757.55 kJ/kWh vs PG-02 <= 5,860.00 (margin +1.75 %)
OK	CW temperature rise 7.00 K vs PG-13 <= 7.00 (margin +0.00 %)
OK	guarantee cover: at IG-01/IG-02 limits and this HMB's auxiliaries the net plant is 602.8 MW / 5,852 kJ/kWh vs Owner PG-01 600 MW / PG-02 5,860 kJ/kWh

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-100
CASE	Site Reference Conditions, natural gas, base load (guarantee)
DOCUMENT	ALP-HMB-SRC-NG-100
STATUS	preliminary - generated by engine hmb v1.0.0
DATA	745a1d896d59ed4c 2026-09-27
NOTE	all values calculated from the database; do not edit